**FPGA 레지스터 Map (Version 2.41)**

| IO | Word Offset | Addr. Offset | Read / Write | Description |
|---|---|---|---|---|
| UARTA | 0x000 | 0x0200_0000 | R/W | UART related registers |
| | ~ | ~ | | |
| | 0x007 | 0x0200_001C | R/W | |

| Bit[31:8] | Bit[7:0] |
|---|---|
| N.A. | UARTREG |

| | |
|---|---|
| UARTREG | 8 bit registers |

| IO | Word Offset | Addr. Offset | Read / Write | Description |
|---|---|---|---|---|
| DOUT0 | 0x010 | 0x0200_0040 | R/W | DOUT0 Port High/Low |
| DOUT1 | 0x011 | 0x0200_0044 | R/W | DOUT1 Port High/Low |
| DOUT2 | 0x012 | 0x0200_0048 | R/W | DOUT2 Port High/Low |

| Bit[31:1] | Bit0 |
|---|---|
| N.A. | H/L |

| | |
|---|---|
| H/L | 0: Port Output Low<br>1: Port Output High |

| IO | Word Offset | Addr. Offset | Read / Write | Description |
|---|---|---|---|---|
| FPGALED | 0x013 | 0x0200_004C | R/W | FPGA LED On/Off |

| Bit[31:1] | Bit0 |
|---|---|
| N.A. | On/Off |

| | |
|---|---|
| On/Off | 0: FPGA LED Off<br>1: FPGA LED On |

| IO | Word Offset | Addr. Offset | Read / Write | Description |
|---|---|---|---|---|
| DC_DRV_EN | 0x020 | 0x0200_0080 | R/W | PWM Driver En/Dis Control |

| Bit[31:1] | Bit0 |
|---|---|
| N.A. | En/Dis |

| | |
|---|---|
| En/Dis | 0: DC Motor PWM Driver Disable<br>1: DC Motor PWM Driver Enable |

<table>
<tr><td>IO</td><td>Word Offset</td><td>Addr. Offset</td><td>Read / Write</td><td colspan="2">Description</td></tr>
<tr><td>DC_PWM_R</td><td>0x021</td><td>0x0200_0084</td><td>R/W</td><td colspan="2">DC Motor right Duty ratio setting</td></tr>
<tr><td>DC_PWM_L</td><td>0x022</td><td>0x0200_0088</td><td>R/W</td><td colspan="2">DC Motor left Duty ratio setting</td></tr>
<tr><td colspan="5">Bit[31:12]</td><td>Bit[11:0]</td></tr>
<tr><td colspan="5">N.A.</td><td>Duty</td></tr>
<tr><td>Duty</td><td colspan="5">PWM duty ratio setting<br>PWM 주파수는 대략 20 kHz<br>0x000 : 0% PWM<br>0x800 : 50% PWM<br>0xFFF : 100% PWM</td></tr>
</table>

<table>
<tr><td>IO</td><td>Word Offset</td><td>Addr. Offset</td><td>Read / Write</td><td colspan="2">Description</td></tr>
<tr><td>DC_ENCCNT_CLR</td><td>0x023</td><td>0x0200_008C</td><td>W</td><td colspan="2">Encoder counter clear</td></tr>
<tr><td colspan="5">Bit[31:1]</td><td>Bit0</td></tr>
<tr><td colspan="5">N.A.</td><td>ECC</td></tr>
<tr><td>ECC</td><td colspan="5">Encoder counter clear<br>All encoder position counters are cleared to zero by writing any value to bit0.</td></tr>
</table>

<table>
<tr><td>IO</td><td>Word Offset</td><td>Addr. Offset</td><td>Read / Write</td><td colspan="2">Description</td></tr>
<tr><td>DC_ENCCNT_MODE</td><td>0x024</td><td>0x0200_0090</td><td>R/W</td><td colspan="2">Encoder counter mode</td></tr>
<tr><td colspan="5">Bit[31:2]</td><td>Bit[1:0]</td></tr>
<tr><td colspan="5">N.A.</td><td>MODE</td></tr>
<tr><td>MODE</td><td colspan="5">Encoder counter mode<br>2’b00: 1X, 2’b01/2’b10: 2X, 2’b11: 4X</td></tr>
</table>

<table>
<tr><td>IO</td><td>Word Offset</td><td>Addr. Offset</td><td>Read / Write</td><td colspan="2">Description</td></tr>
<tr><td>DC_ENCST_R</td><td>0x028</td><td>0x0200_00A0</td><td>R</td><td colspan="2">Right encoder pulse status</td></tr>
<tr><td>DC_ENCST_L</td><td>0x029</td><td>0x0200_00A4</td><td>R</td><td colspan="2">Left encoder pulse status</td></tr>
<tr><td colspan="5">Bit[31:2]</td><td>Bit[1:0]</td></tr>
<tr><td colspan="5">N.A.</td><td>ENCPHA</td></tr>
<tr><td>ENCPHA</td><td colspan="5">ENCPHA[1:0] = {B, A}</td></tr>
</table>

| IO | Word Offset | Addr. Offset | Read / Write | Description |
|---|---|---|---|---|
| DC_ENC_POSCNT_R | 0x02A | 0x0200_00A8 | R | Right Encoder Position Counter |
| DC_ENC_POSCNT_L | 0x02B | 0x0200_00AC | R | Left Encoder Position Counter |

| Bit[31:0] |
|---|
| EncPos |

| | |
|---|---|
| EncPos | Encoder Counter Value(32 bit 2’s complement value)<br>32’h8000_0000 ~ 32’h7FFF_FFFF |

| IO | Word Offset | Addr. Offset | Read / Write | Description |
|---|---|---|---|---|
| DC_ENC_VELCNT_R | 0x02C | 0x0200_00B0 | R | Right Encoder Velocity Counter0 |
| DC_ENC_VELCNT_L | 0x02D | 0x0200_00B4 | R | Left Encoder Velocity Counter1 |

| Bit31 | Bit[30:0] |
|---|---|
| Dir | EncVel |

| | |
|---|---|
| Dir | 0: Counterclockwise<br>1: Clockwise |
| EncVel | 40MHz의 CPU clock을 1/8로 낮춘 클럭으로 Encoder A 신호의 edge와 edge 사이의 시간을 counting<br>31’h7FFF_FFFF에서 saturation됨. |

| IO | Word Offset | Addr. Offset | Read / Write | Description |
|---|---|---|---|---|
| STEP_PHASE_R | 0x030 | 0x0200_00C0 | R/W | Right stepping motor phase |
| STEP_PHASE_L | 0x031 | 0x0200_00C4 | R/W | Left stepping motor phase |

| Bit[31:8] | Bit[3:0] |
|---|---|
| N.A. | STPPHA |

| | |
|---|---|
| STPPHA | Stepmotor step phase<br>STPPHA[3:0] = {B, /B, A, /A} |

| IO | Word Offset | Addr. Offset | Read / Write | Description |
|---|---|---|---|---|
| BLDC_DRV_MODE | 0x040 | 0x0200_0100 | R/W | BLDC commutation/drive mode |

| Bit[31:1] | Bit0 |
|---|---|
| N.A. | MODE |

| | |
|---|---|
| MODE | 0: BLDC commutation by FW<br>1: BLDC commutation by HW |

| IO | Word Offset | Addr. Offset | Read / Write | Description |
|---|---|---|---|---|
| BLDC_DRV_EN | 0x041 | 0x0200_0104 | R/W | BLDC Driver En/Dis Control |

| Bit[31:1] | Bit0 |
|---|---|
| N.A. | En/Dis |

| | |
|---|---|
| En/Dis | 0: BLDC Motor PWM Driver Disable<br>1: BLDC Motor PWM Driver Enable |

| IO | Word Offset | Addr. Offset | Read / Write | Description |
|---|---|---|---|---|
| BLDC_PWM0 | 0x042 | 0x0200_0108 | R/W | BLDC Motor ‘A Phase’ Duty ratio setting |
| BLDC_PWM1 | 0x043 | 0x0200_010C | R/W | BLDC Motor ‘B Phase’ Duty ratio setting |
| BLDC_PWM2 | 0x044 | 0x0200_0110 | R/W | BLDC Motor ‘C Phase’ Duty ratio setting |

| Bit[31:12] | Bit[11:0] |
|---|---|
| N.A. | Duty |

| | |
|---|---|
| Duty | PWM 주기는 대략 20 kHz<br>0x000 : 0% PWM<br>0x800 : 50% PWM<br>0xFFF : 100% PWM |

| IO | Word Offset | Addr. Offset | Read / Write | Description |
|---|---|---|---|---|
| BLDC_HALLCNT_CLR | 0x045 | 0x0200_0114 | W | BLDC hall sensor counter clear |

| Bit[31:1] | Bit0 |
|---|---|
| N.A. | HCC |

| | |
|---|---|
| HCC | Encoder counter clear<br>All hall sensor position counters are cleared to zero by writing any value to bit0. |

| IO | Word Offset | Addr. Offset | Read / Write | Description |
|---|---|---|---|---|
| BLDC_DRIVE | 0x046 | 0x0200_0118 | R/W | BLDC motor HW commutation drive duty value |

| Bit[31:12] | Bit[11:0] |
|---|---|
| N.A. | Duty |

| | |
|---|---|
| Duty | PWM 주기는 대략 20 kHz<br>0x000 : 0% PWM<br>0x800 : 50% PWM<br>0xFFF : 100% PWM |

| IO | Word Offset | Addr. Offset | Read / Write | Description |
|---|---|---|---|---|
| BLDC_HALLST | 0x048 | 0x0200_0120 | R | BLDC Hall Sensor Phase In |

| Bit[31:3] | Bit[2:0] |
|---|---|
| N.A. | HallPha |

| | |
|---|---|
| HallPha | BLDC 모터의 3상 Hall sensor phase 상태 read<br>HallPha[2:0] = {C, B, A} |

| IO | Word Offset | Addr. Offset | Read / Write | Description |
|---|---|---|---|---|
| BLDC_POSCNT | 0x049 | 0x0200_0124 | R | BLDC Hall sensor position counter |

| Bit[31:0] |
|---|
| HallPos |

| | |
|---|---|
| HallPos | Hall counter value(32 bit 2's complement value)<br>32'h8000_0000 ~ 32'h7FFF_FFFF |

| IO | Word Offset | Addr. Offset | Read / Write | Description |
|---|---|---|---|---|
| BLDC_VELCNT | 0x04A | 0x0200_0128 | R | BLDC Hall sensor velocity counter |

| Bit31 | Bit[30:0] |
|---|---|
| Dir | HallVel |

| | |
|---|---|
| Dir | 0: Counterclockwise<br>1: Clockwise |
| HallVel | 40MHz의 CPU clock을 1/8로 낮춘 클럭으로 Hall sensor A 신호의 edge와 edge 사이의 시간을 counting<br>31'h7FFF_FFFF에서 saturation됨. |

| IO | Word Offset | Addr. Offset | Read / Write | Description |
|---|---|---|---|---|
| MUX_SEL | 0x050 | 0x0200_0140 | R/W | Analog Mux Selection Control |

| Bit[31:3] | Bit[2:0] |
|---|---|
| N.A. | SEL |

| | |
|---|---|
| SEL | Mux Channel Num connected to Mux Output |

| IO | Word Offset | Addr. Offset | Read / Write | Description |
|---|---|---|---|---|
| ADCIN | 0x051 | 0x0200_0144 | R | 16 bit ADC Conversion Result Read |

| Bit[31:16] | Bit[15:0] |
|---|---|
| 16'h0000 | ADCVal |

| | |
|---|---|
| ADCVal | 0x0000: -5.0V<br>0x8000: 0.0V<br>0xFFFF: 5.0V |

| IO | Word Offset | Addr. Offset | Read / Write | Description |
|---|---|---|---|---|
| DAC0 | 0x058 | 0x0200_0160 | W | 16 bit DAC Channel #0 Out |
| DAC1 | 0x059 | 0x0200_0164 | W | 16 bit DAC Channel #1 Out |
| DAC2 | 0x05A | 0x0200_0168 | W | 16 bit DAC Channel #2 Out |
| DAC3 | 0x05B | 0x0200_016C | W | 16 bit DAC Channel #3 Out |
| DAC4 | 0x05C | 0x0200_0170 | W | 16 bit DAC Channel #4 Out |
| DAC5 | 0x05D | 0x0200_0174 | W | 16 bit DAC Channel #5 Out |
| DAC6 | 0x05E | 0x0200_0178 | W | 16 bit DAC Channel #6 Out |
| DAC7 | 0x05F | 0x0200_017C | W | 16 bit DAC Channel #7 Out |

| Bit[31:16] | Bit[15:0] |
|---|---|
| N.A. | DACVal |

| | |
|---|---|
| DACVal | 0x0000: -5.0V<br>0x8000: 0.0V<br>0xFFFF: 5.0V |

| IO | Word Offset | Addr. Offset | Read / Write | Description |
|---|---|---|---|---|
| FPGAVER | 0x0FF | 0x0200_03FC | R | FPGA 버전 확인 |

| Bit[31:16] | Bit[15:0] |
|---|---|
| 16’h0000 | Ver |

| | |
|---|---|
| Ver | 4bit 단위로 하나의 hexa 값을 표현<br>총 4자리의 version을 표현 |